

20 Phase change for Q_r (content warning: #explosion)

Why are we learning this?

In previous lectures, we studied reactor energy balances and learned how heat transfer can be used to control reactor temperature. Today we introduce another powerful mechanism for removing energy from a reactor: Phase Change

Phase changes such as vaporization can remove large amounts of energy because they involve latent heat. This concept is important in:

- Reactor safety
- Runaway-reaction prevention
- Pressure-relief systems
- Reactive distillation
- Process design

By the end of today's lesson, you should understand:

- Why vaporization can remove large amounts of energy.
- How rupture disks can provide emergency cooling.
- How latent heat appears in reactor energy balances.
- Why phase change can help prevent thermal runaway.
- How phase change can be incorporated intentionally into reactor design.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress

(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws
(Kinetics)



Equilibrium
(Reversible Reactions)



Equilibrium Conversion
(Thermodynamic Limitations)



One Reaction: Solving in terms of Reaction Progress



One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)



Catalytic Reactions



Numerical Methods
(Python)



► Energy Balances ◀



One Reaction: Non-Isothermal Reactor Design



Multiple Reactions



Reactor Safety



Challenging the Well-Mixed Assumption
(Residence Time Distributions)



Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Big Idea

Boiling removes energy. Lots of energy. Latent heat associated with phase change can provide an additional mechanism for reactor cooling.

We saw when we studied the Monsanto explosion that the main safety feature on the reactor was a rupture disk that was set to burst when the pressure in the reactor reached 700 psi.

Why Study a Reactor Accident?

Reactor accidents often reveal important engineering principles. In this example, a pressure-relief system was intended not only to reduce pressure but also to remove energy from the reactor. Understanding these mechanisms helps engineers design safer reactors and prevent thermal runaway.

This [video](#) illustrates how a rupture disk works (it is an animation of a pipe fitted with a rupture disk and it has no sound).

What Is a Rupture Disk?

A rupture disk is a safety device designed to fail at a specified pressure.

Pressure Too High



Disk Ruptures



Material Escapes



Pressure Drops

Its purpose is to prevent catastrophic equipment failure caused by excessive pressure.

According to Fogler, the rupture disk in the Monsanto reactor was fitted with a relief valve with a 2 inch orifice that vented to the atmosphere. The maximum flowrate through the relief valve (from fluid dynamics, calculated for us by Fogler) was hence 830 kg/min.

If the rupture disk had worked properly, once it broke, the pressure in the reactor would have dropped, causing the water to vaporize. The water vapor would've then exited the reactor through the relief valve.

The maximum rate of energy removal (Q_r) due to this process would be:

Pressure Drop Can Cause Vaporization

When pressure decreases: Boiling Point Decreases

A liquid that was stable at high pressure may suddenly vaporize. Vaporization requires energy:

Liquid + Energy $\rightarrow$ Vapor

The required energy comes from the reactor contents, cooling the system.

Latent Heat Is Large

Phase changes involve latent heat rather than sensible heat. Latent heat can remove large amounts of energy while temperature remains nearly constant.

Heating a Liquid $\neq$ Boiling a Liquid

Boiling typically removes much more energy per unit mass. This makes vaporization a powerful cooling mechanism.

Phase Change Can Be a Design Feature

Phase change is not only a safety mechanism. It can also be incorporated intentionally into reactor operation. Examples include:

- Solvent boiling and condensation loops.
- Evaporative cooling.
- Reactive distillation.
- Feed vaporization systems.

Engineers often exploit phase change because of its exceptional cooling capacity.

One can imagine designing a reactor where, for example, the solvent boils off continuously, is condensed, and then is added back to the reactor:

The energy balance for this process is:

Where the heat of vaporization provides an additional heat removal term.

A New Energy-Removal Term

In previous energy balances, cooling was often represented through heat-transfer terms such as:

$$UA(T_a - T)$$

With phase change, an additional term appears:

$$\text{Mass Vaporized} \times \text{Heat of Vaporization}$$

This represents energy removed through latent heat rather than conventional heat transfer.

Alternatively, one might design a reactor to boil off a product species as a means of distillation. This is called reactive distillation; you can find a short and informative video of reactive distillation [here](#).

Reactive Distillation

Reactive distillation combines:

Reaction + Separation

within a single unit operation.

As products are formed, they are simultaneously removed by vaporization. This can:

- Improve separation efficiency.
- Shift equilibrium.
- Improve process economics.

Another way to incorporate phase change is to introduce a *liquid* feed to the reactor, which then vaporizes in the reactor due to absorbing the heat generated by the reaction. For example:

The energy balance for this example is:

Where here, the main difference between the energy balance that we've been working with is that instead of accounting for sensible heat changes between the feed and reactor, we are now accounting for latent heat changes.

Sensible Heat and Latent Heat Are Different

Sensible Heat: Energy associated with changing temperature.

Same Phase, Different Temperature

Latent Heat: Energy associated with changing phase.

Same Temperature, Different Phase

Phase-change systems often require latent-heat terms in addition to conventional sensible-heat terms.

Summary and Key Takeaways

Today we explored how phase change can contribute to energy removal in chemical reactors.

Key Ideas

- ✓ Vaporization can remove large amounts of energy through latent heat.
- ✓ Pressure relief systems may provide both pressure control and cooling.
- ✓ Depressurization can trigger vaporization and associated energy removal.
- ✓ Latent heat provides an additional term in reactor energy balances.
- ✓ Engineers can intentionally incorporate phase change into reactor designs.
- ✓ Reactive distillation combines reaction and separation within a single unit operation.

Most Important Idea

Phase Change



Latent Heat Removal



Temperature Control



Improved Reactor Safety

Phase change is not only a thermodynamic phenomenon—it can be a powerful reactor-design and reactor-safety tool.